GLOBAL CHAOTICITY IN THE PYTHAGOREAN THREE-BODY PROBLEM

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Abstract. The effects of small changes in the initial conditions of the Pythagorean three-body problem are investigated by computer simulations. This problem consists of three interacting bodies with masses 3, 4 and 5 placed with zero velocities at the apices of a triangle with sides 3, 4 and 5. The final outcome of this motion is that two bodies form a binary and the third body escapes. We attempt to establish regions of the initial positions which give regular and chaotic motions. The vicinity of a small neighbourhood around the standard initial position of each body defines a regular region. Other regular regions also exist. Inside these regions the parameters of the triple systems describing the final outcome change continuously with the initial positions. Outside the regular regions the variations of the parameters are abrupt when the initial conditions change smoothly. Escape takes place after a close triple approach which is very sensitive to the initial conditions. Time-reversed solutions are employed to ensure reliable numerical results and distinguish between predictable and non-predictable motions. Close triple approaches often result in non-predictability, even when using regularization; this introduces fundamental difficulties in establishing chaotic regions.

Key words: Three-body problem, chaotic motions, regularization.

1. Introduction

The general dynamical behaviour of three bodies with negative total energy can be described as follows. At the beginning there is interplay interspersed by ejections. Eventually this leads to a close triple approach and the formation of a binary. If the energy of the binary exceeds the total energy in absolute value, then the third body acquires a hyperbolic orbit with respect to the binary and we have the condition for escape. Thus the highly chaotic motion is followed by an orderly behaviour as time goes to infinity. This behaviour is well established in the literature for a great variety of triple systems (Anosova, 1969; Szebehely, 1973; Standish, 1976; Agekian and Anosova, 1977; Anosova, 1986; Anosova and Zavalov, 1989; Johnstone and Rucinski, 1991).

The present paper is concerned with a more global effort which seeks to establish regions of initial conditions leading to chaoticity in the final outcome. In other

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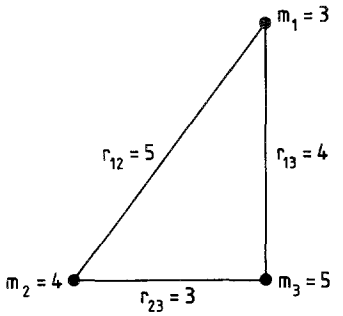


Fig. 1. Initial positions of the Pythagorean three-body problem.

words, regions of initial conditions are found where the final outcome shows high sensitivity to small changes of the initial conditions. Since all the initial velocities are zero, our two-dimensional system is represented initially by a six-vector $\mathbf{r_o}(x_1, y_1; x_2, y_2; x_3, y_3)$. The final outcome can be described by identifying the mass of the escaping single body (m_1, m_2, m_3), the direction of escape (β) with respect to the x-axis, the time when the escape condition is satisfied, and the orbital elements of the final binary. In fact, the first two of these parameters – the identity of the escaping mass and the escape angle – can be considered as the essential parameters of the final outcome.

When small changes in $\mathbf{r_o}$ result in high sensitivity in β or in qualitative changes in the identity of the escaping single body, we refer to this behaviour as *globally* chaotic. It is essential to distinguish between global and local chaoticity in this problem since the initial motion is always chaotic when the three bodies exhibit interplay (excepting measure zero sets for periodic orbits), and the final motion is always regular. In this paper we are concerned with global chaoticity; i.e. the sensitivity of the final outcome due to changes in the initial motions.

The example selected in this paper is the so-called Pythagorean three-body problem because this configuration has received considerable attention ever since Meissel (1893) and Burrau (1913), although the complete solution was obtained much later (Szebehely and Peters, 1967). The three participating masses are defined by $m_1 = 3$, $m_2 = 4$ and $m_3 = 5$. As shown in Figure 1, these bodies are initially located at the apices of a triangle with sides $r_{12} = 5$, $r_{23} = 3$ and $r_{31} = 4$. Since all the initial velocities are always zero in this paper, we are dealing with two-dimensional motions of a system having negative total energy.

2. Geometry of the Manifolds of the Initial Conditions

The regions of chaoticity and regularity are established by keeping the locations of two masses fixed and changing the initial position of the third body. Although more general types of changes exist, in this first attempt to establish regions of chaoticity we only consider the restricted scheme of varying one body at a time.

First we keep m_2 and m_3 fixed at their original (Pythagorean) positions and vary the initial position of m_1. Then m_1 and m_3 are kept fixed at their original positions and m_2 is varied, and finally the position of m_3 is varied. The sets of initial conditions are given by

$$x_1 = 1 + \delta_1 cos\alpha_1, \quad x_2 = -2 + \delta_2 cos\alpha_2, \quad x_3 = 1 + \delta_3 cos\alpha_3,$$

$$y_1 = 3 + \delta_1 sin\alpha_1, \quad y_2 = -1 + \delta_2 sin\alpha_2, \quad y_3 = -1 + \delta_3 sin\alpha_3, \qquad (1)$$

Here $\delta_i = 5 \times 10^{-k}$ with $k = 8, 7, ..., 3$, whereas $\alpha_i = 3.6n$, with $n = 0, 1, 2, ..., 99$, and $i = 1, 2, 3$.

The first set of initial conditions is given by $\delta_2 = \delta_3 = 0$ and $\alpha_1 = 0$. We refer to this simulation as experiment number $N = 1$. Experiment number 2 increases α_1 to 3.6 degrees, keeping all the other parameters fixed. Thus we select 100 points on a circle of radius $\delta_1 = 5 \times 10^{-8}$ around m_1 and perform 100 integrations. The next set of experiments is obtained when δ_1 is changed to 5×10^{-7}. In total there are 600 experiments changing the initial position of m_1. This is followed by 600 experiments when the positions of m_1 and m_3 are not changed and m_2 is placed initially on six different circles (drawn around its original position) at 100 different angles (α_2), and finally the same is done for m_3.

3. Description of the Numerical Technique

We employ a three-body regularization method (Aarseth and Zare, 1974; Aarseth, 1976) for the numerical integrations. This procedure enables not only double approaches but also triple approaches to be studied with increased confidence, although interactions of long duration are still notoriously difficult. Each example is first calculated with a standard tolerance of 1×10^{-13}, using the Bulirsch-Stoer (1966) integrator. The calculation is repeated with an improved tolerance of 1×10^{-15} if the time-reversed solution does not satisfy the specified accuracy criterion; this is close to the practical lower limit for double precision (i.e. 16 figures) calculation.

The calculation is continued until one body escapes, using the escape criterion of Standish (1971) for the relative velocity outside a characteristic distance $3R_{\rm grav}$, where $R_{\rm grav} = 3.67$ is the gravitational radius when the constant of gravity is unity. (The corresponding standard crossing time is $T_{cr} = 3.84$). Time-reversed solutions are then obtained in order to compare the results with the initial conditions. This accuracy control is more reliable than the conventional energy check which

is merely a necessary condition and may be misleading. In order to minimize numerical problems due to long life-times, we have also introduced a criterion of conditional escape for distant ejections (Anosova, 1986) for separations exceeding $5R_{\rm grav}$. In this way we can distinguish between 'true' and 'conditional' escape. For the same reason, we employ an upper time limit of 200 time units; such cases are referred to as incomplete experiments.

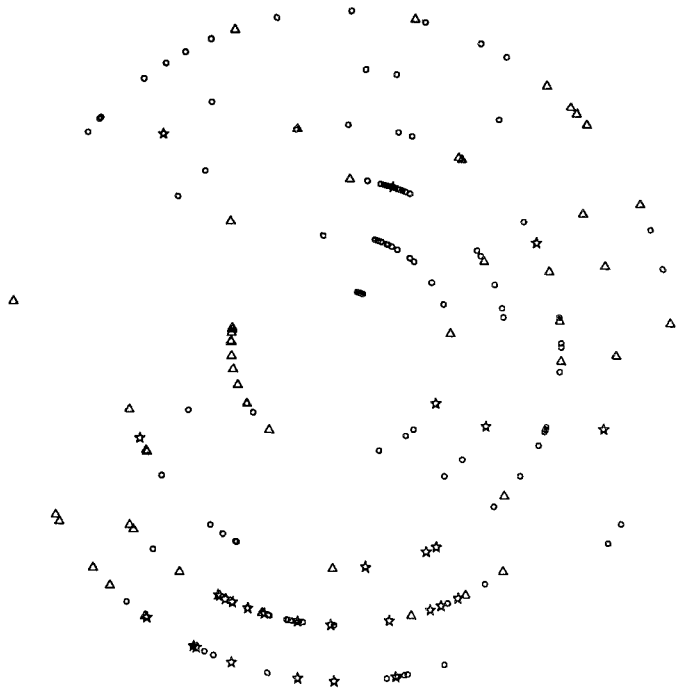
In our analysis we only include experiments where the r.m.s. deviations between the initial and final coordinates and velocities, evaluated separately, are < 0.001 in scaled units. This procedure has been adopted because the equilibrium r.m.s. velocity is within a factor $\simeq 3$ of $R_{\rm grav}$ in our calculation units. The cases with acceptable errors are termed 'predictable' ones. Cases with deviations larger than 0.001 are excluded from further study, and are called 'non-predictable'.

4. Stochasticity and Regularity

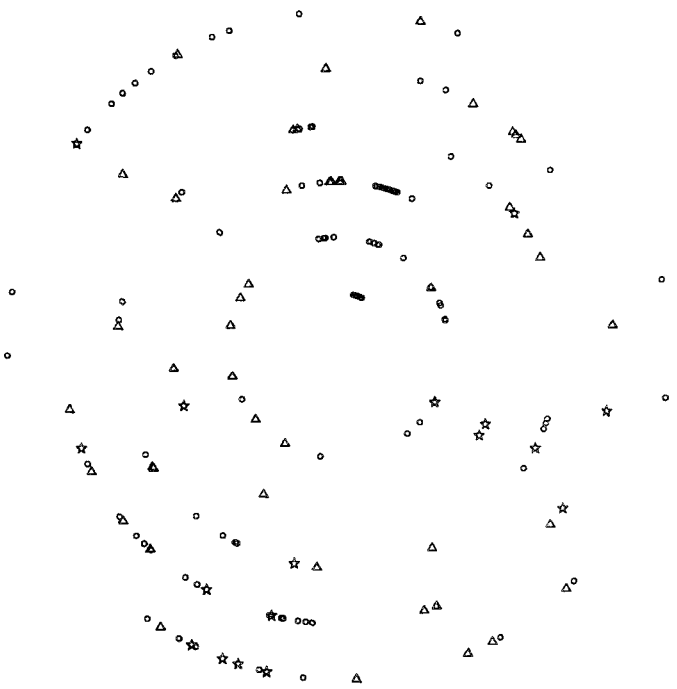
The main results are presented in Figures 2–4. Figure 2 shows the distributions of the escape angles β for systems with initial positions of the bodies along the circles for different δ_i. Here the innermost circle corresponds to $\delta_i = 5.0 \times 10^{-8}$ and the outermost ring to $\delta_i = 5.0 \times 10^{-3}$. We see some regions of clustering for β; e.g., around $\beta = 71.4$ degrees, corresponding to the original Pythagorean triangle. On the other hand, there are also some gaps. The latter may be due to either the absence of escape in these directions or to non-predictable motions. In Figure 2 circles denote escape of the component m_1, triangles correspond to escape of m_2, and asterisks indicate escape of the heavy body m_3.

The regions of clustering of β may correspond statistically to regions of regularity (or stability) of the motions in the three-body problem. Note that the regular motions have escape times within the interval [45, 120] scaled time units, with a mean escape time of 65 (or 17 crossing times). On the other hand, the non-predictable regions correspond to somewhat longer escape times; typically 60 – 200 units or more. The latter may be due to (i) a mathematical instability (i.e. numerical errors), and/or (ii) a strong dynamical instability (or chaos) of the motions in relation to small variations in initial conditions, with subsequent magnification during close approaches.

We have no simple way of separating these two effects at present. It should be noted that this work has employed an efficient three-body regularization method combined with an accurate integrator. However, since the non-predictable examples tend to have longer escape times than the predictable ones, we would also expect larger numerical errors on average for the former since neighbouring trajectories are subject to exponential deviations (Dejonghe and Hut, 1986). Anosova (1990) has shown that error accumulation in the general three-body problem is strongly correlated with the nature of the motions, and the uncertainties of the solutions are complicated functions of evolution time as well as separations. Thus we would only expect a small reduction in the fraction of non-predictable examples if integrations



(a)



(b)

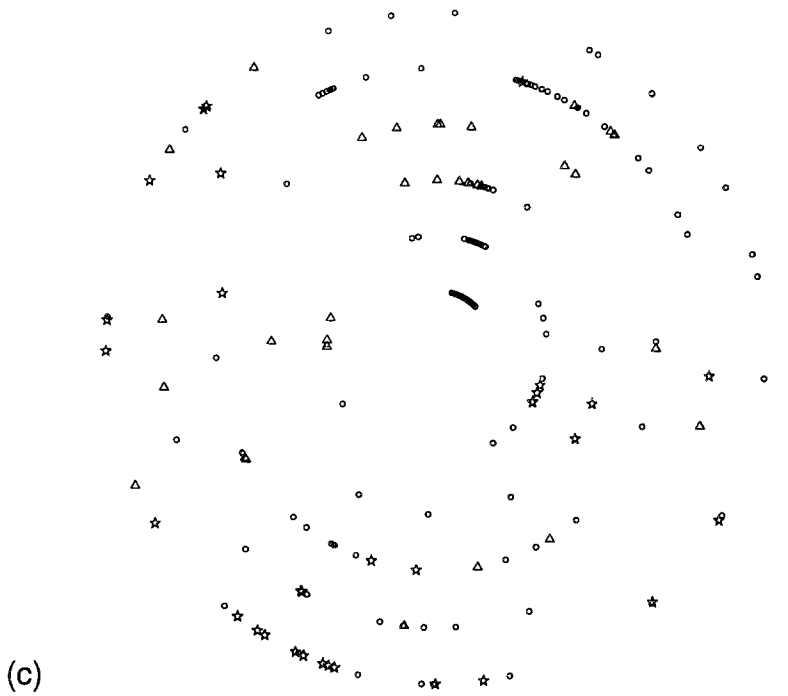
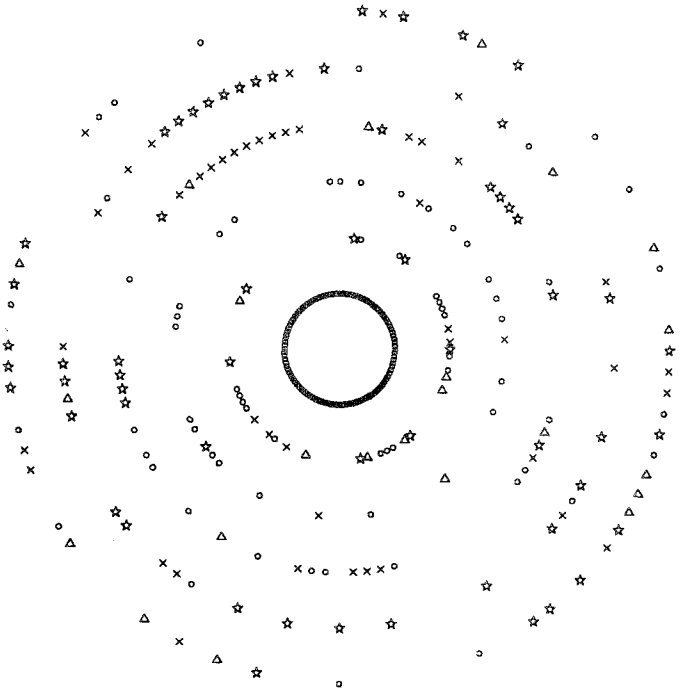


Fig. 2. Distributions of the angles of escape β for different values of initial deviations. The corresponding values of δ_i from the innermost to the outermost circle are 5.0×10^{-8}, 5.0×10^{-7}, 5.0×10^{-6}, 5.0×10^{-5}, 5.0×10^{-4} and 5.0×10^{-3}. Results for deviations around m_1, m_2 and m_3 are shown in (a), (b) and (c), respectively. Escape of components m_1, m_2 or m_3 is denoted by circles, triangles and asterisks. The x-axis is horizontal.

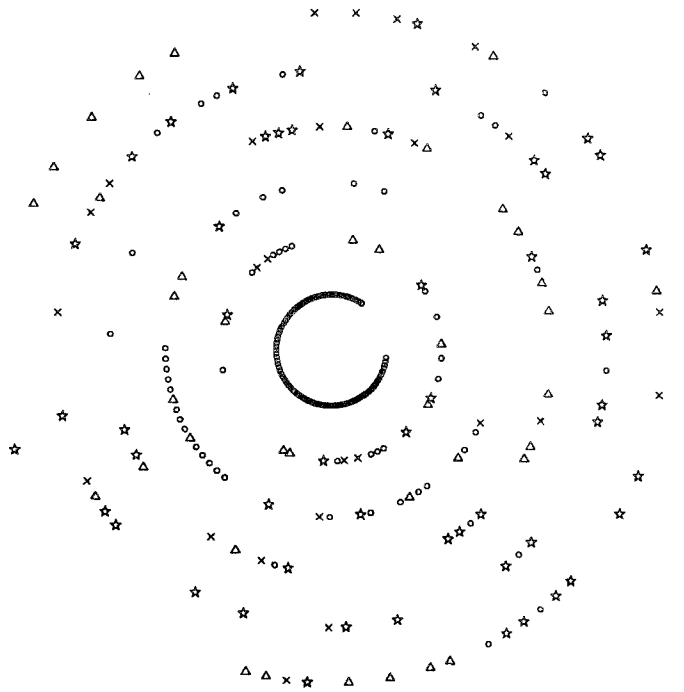
were performed at higher accuracy (i.e. quadruple precision).

In Figure 3 we show the dependence of the escape angles β on the angle α defining the initial phase angle of the relevant body. In these figures circles correspond to β from 0 to 90 degrees; triangles represent β from 90 to 180; asterisks, β from 180 to 270; crosses, β from 270 to 360 degrees. There are regions containing the same symbol of β. These regions correspond to a crossing of the regular areas by the circles δ_i. In case of variation of body m_1 (Figure 3a) we have a continuous closed ring at $\delta_i = 5.0 \times 10^{-8}$, where the angles β are close to 70 degrees. Other regular regions are also shown in Figure 3a; two sequences of crosses and asterisks are seen in the upper left corner (α changes from about 90 to 130 degrees); two small regions of asterisks are observed elsewhere (in directions to $\beta = 45$ and 180 degrees). In Figures 3b and 3c the central rings are incomplete; they have a 'horseshoe' form. In addition, some elongated regions exist where the angles β are very similar.

In Figure 4 the dependence $T(\alpha)$ is displayed. Here T is the escape time; the moment when either the Standish criterion or the conditional escape criterion is



(a)



(b)

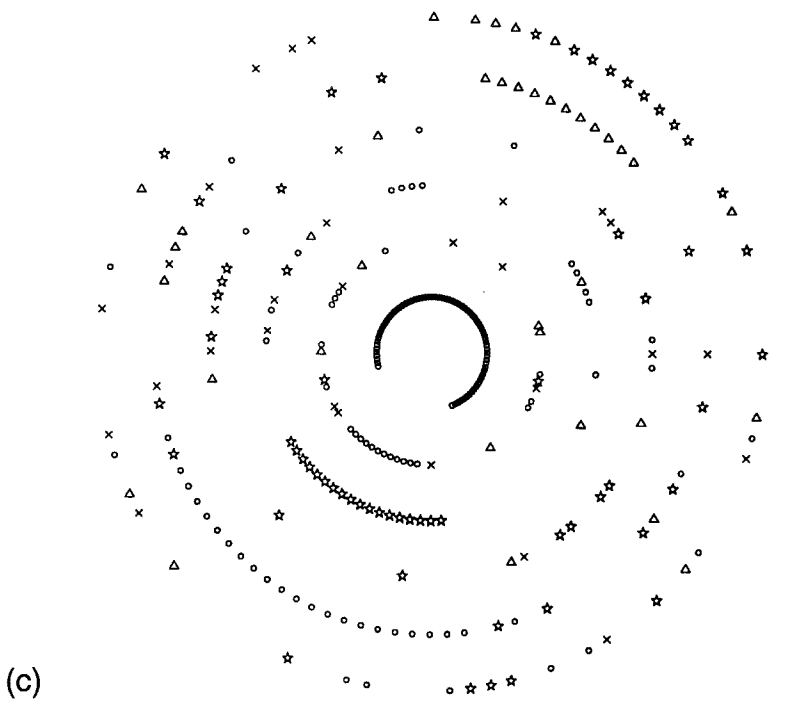
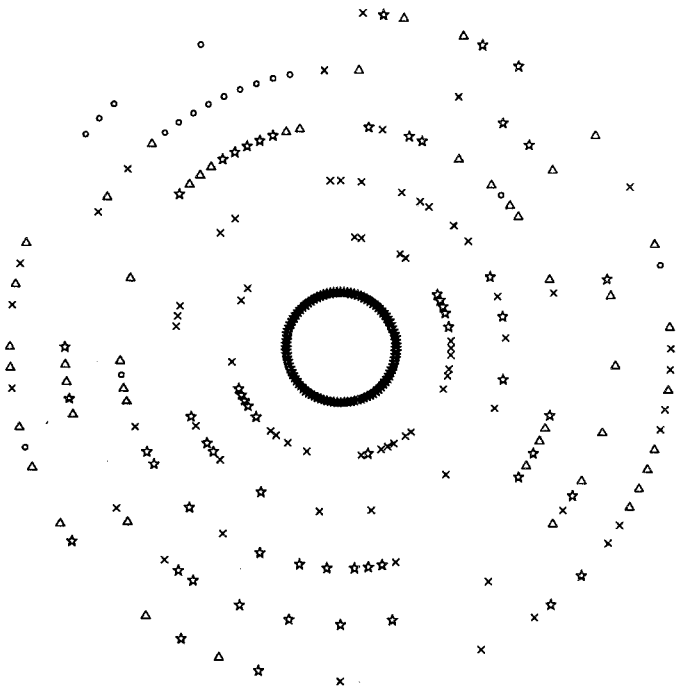


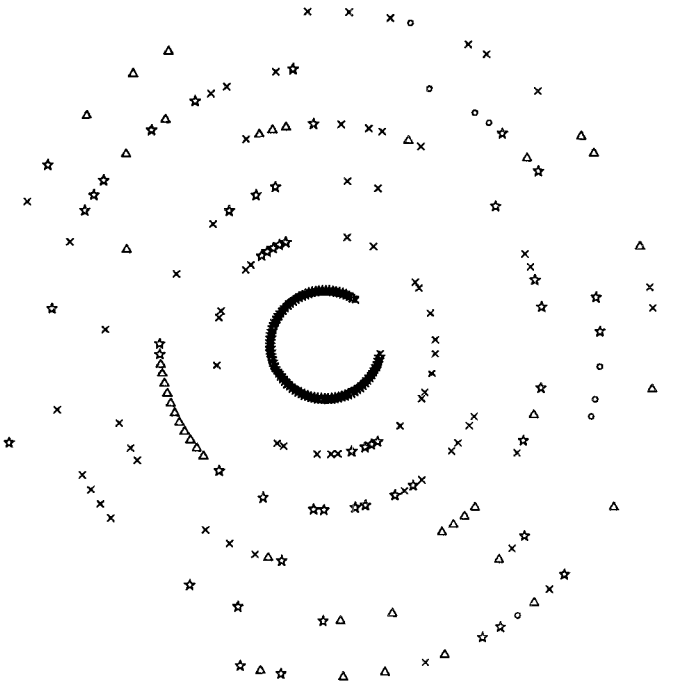
Fig. 3. Dependence of the escape angle β on the initial azimuth angle α. Symbols are defined in the text. Variations around m_1, m_2 and m_3 are shown in (a), (b) and (c).

satisfied. The following symbols are adopted: circles correspond to the quantity T within the interval from 40 to 50 time units; triangles, to the interval from 50 to 60 units; asterisks, from 60 to 70 units; and crosses show the cases with T exceeding 70 time units. The voids in Figures 3 and 4 denote the non-predictable examples. The regions of regular behaviour of β coincide with similar regions for T.

We have also scanned three regular regions with a smaller step, chosen as 0.36 degrees. The first region includes the central point and elongates along the straight line connecting the bodies m_1 and m_2. For each value of α we select three radial displacements $\delta_i = 3.0 \times 10^{-7}$, 5.0×10^{-7} and 7.0×10^{-7} within sectors with a central angle equal to 36 degrees. The fine structure of the regular regions shows some large oscillations of T at the boundaries of regular regions. These boundaries correspond to conditional escape (Agekian and Anosova, 1977; Anosova, 1986) when very distant ejection without escape takes place. Small differences in the ejection interval may give rise to large variations of the result at the next triple approach, and the largest instability takes place just at the boundaries of the regular regions. This is the main reason for the onset of chaos. In these cases we have 'non-predictable' motion.



(a)



(b)

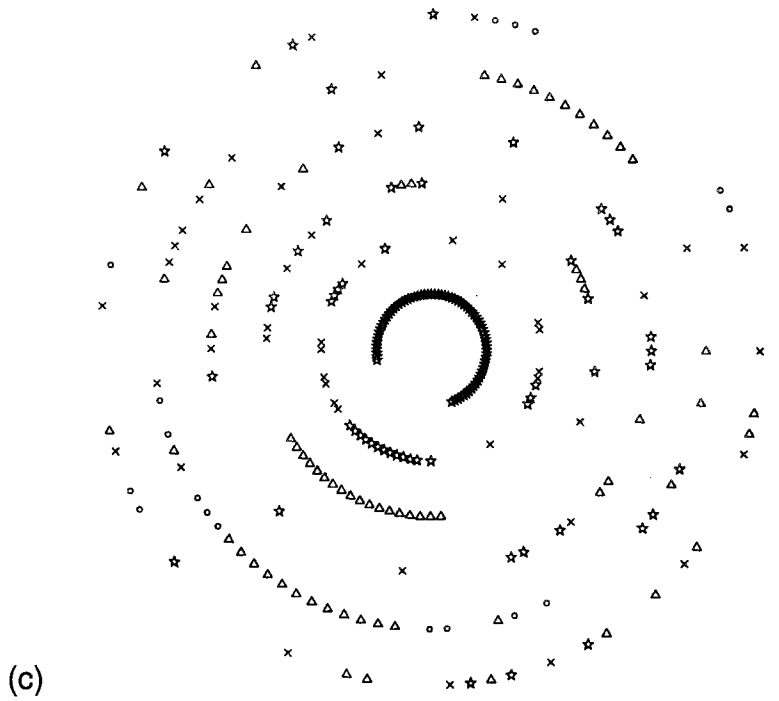


Fig. 4. Dependence of the escape time T on α. Explanations are given in the text. Variations around m_1, m_2 and m_3 are shown in (a), (b) and (c).

TABLE I

Escape times of triple systems.

δ	N_1	$< T_1 >$	σ_1	N_2	$< T_2 >$	σ_2	N_3	$< T_3 >$	σ_3
5×10^{-3}	47	64.8	2.4	35	65.9	2.6	46	58.2	2.6
5×10^{-4}	42	62.1	2.1	37	65.8	2.5	55	57.8	1.8
5×10^{-5}	47	62.3	1.2	34	72.7	2.7	30	64.9	1.8
5×10^{-6}	28	79.0	3.1	36	67.0	1.8	38	62.1	1.6
5×10^{-7}	34	71.4	0.8	30	73.3	1.3	35	70.9	1.2
5×10^{-8}	100	66.0	0.2	83	65.4	0.3	74	64.6	0.3

5. Statistical Quantities

Some results for the variation of the bodies m_1, m_2 and m_3 are given in Tables I–VII. These tables contain average quantities and some r.m.s. deviations for nine parameters. The actual number of escapers for each set of experiments (i.e. δ_i and N_1, N_2, N_3) is shown in Columns 2, 5 and 8 of Table I. Here we assume

TABLE II

Escape angles.

δ	$< \beta_1 >$	σ_1	$< \beta_2 >$	σ_2	$< \beta_3 >$	σ_3
5×10^{-3}	172.4	14.1	196.4	14.4	188.0	13.9
5×10^{-4}	235.9	12.3	200.8	15.3	130.6	11.9
5×10^{-5}	205.8	17.9	182.3	17.2	201.1	17.1
5×10^{-6}	96.3	16.4	105.3	11.3	189.5	15.0
5×10^{-7}	143.3	16.8	130.2	18.9	134.8	18.3
5×10^{-8}	71.3	0.2	70.2	0.3	66.8	0.9

TABLE III

Semi-major axes.

δ	$< A_1 >$	σ_1	$< A_2 >$	σ_2	$< A_3 >$	σ_3
5×10^{-3}	0.523	0.025	0.566	0.027	0.442	0.029
5×10^{-4}	0.435	0.022	0.439	0.024	0.616	0.024
5×10^{-5}	0.480	0.024	0.505	0.025	0.490	0.033
5×10^{-6}	0.506	0.027	0.414	0.038	0.573	0.033
5×10^{-7}	0.530	0.023	0.530	0.030	0.510	0.027
5×10^{-8}	0.545	0.015	0.483	0.020	0.414	0.025

that the perimeter of the configuration triangle defining the triple approach is smaller than the virial system size (Agekian and Anosova, 1967). The minimum moment of inertia has been determined, and this epoch has been defined as that of

TABLE IV

Eccentricities.

δ	$< E_1 >$	σ_1	$< E_2 >$	σ_2	$< E_3 >$	σ_3
5×10^{-3}	0.788	0.028	0.825	0.035	0.740	0.028
5×10^{-4}	0.754	0.034	0.739	0.039	0.715	0.023
5×10^{-5}	0.815	0.034	0.761	0.044	0.725	0.049
5×10^{-6}	0.918	0.029	0.931	0.016	0.899	0.025
5×10^{-7}	0.784	0.048	0.864	0.039	0.884	0.033
5×10^{-8}	0.990	0.000	0.991	0.000	0.952	0.011

TABLE V

Minimum perimeters and moments of inertia.

δ	$< P_1 >$	$< P_2 >$	$< P_3 >$	$< I_1 >$	$< I_2 >$	$< I_3 >$
5×10^{-3}	1.36	1.71	1.13	1.13	1.70	0.79
5×10^{-4}	1.12	1.08	1.76	0.65	0.68	1.78
5×10^{-5}	1.51	1.56	1.33	1.08	1.16	0.87
5×10^{-6}	1.63	1.43	1.89	1.51	1.48	1.62
5×10^{-7}	1.50	1.47	1.61	1.29	1.57	1.31
5×10^{-8}	1.88	1.68	1.44	1.50	1.16	1.05

TABLE VI

Minimum two-body distances.

δ	$< R_1 >$	σ_1	$< R_2 >$	σ_2	$< R_3 >$	σ_3
5×10^{-3}	−3.51	0.04	−3.64	0.14	−3.51	0.04
5×10^{-4}	−3.66	0.13	−3.66	0.12	−3.44	0.03
5×10^{-5}	−3.58	0.08	−3.88	0.19	−3.49	0.05
5×10^{-6}	−3.80	0.25	−3.50	0.06	−3.42	0.01
5×10^{-7}	−3.50	0.06	−3.64	0.16	−3.45	0.03
5×10^{-8}	−3.48	0.02	−3.76	0.14	−3.60	0.10

Results are logarithmic to base 10.

the close triple approach. The minimum perimeter before the escape epoch then characterizes the smallest size of the system. Tables I - VII contain average values

TABLE VII

Time reversal errors in coordinates and velocities.

δ	$< \epsilon_1 >$	$< \epsilon_2 >$	$< \epsilon_3 >$	$< \nu_1 >$	$< \nu_2 >$	$< \nu_3 >$
5×10^{-3}	−5.39	−5.56	−5.99	−5.90	−6.08	−6.44
5×10^{-4}	−5.04	−4.71	−6.58	−5.50	−5.17	−7.05
5×10^{-5}	−4.60	−4.26	−4.46	−5.06	−4.72	−4.92
5×10^{-6}	−3.78	−3.79	−4.70	−4.26	−4.25	−5.16
5×10^{-7}	−3.85	−3.72	−3.79	−4.31	−4.18	−4.26
5×10^{-8}	−3.66	−3.80	−3.86	−4.13	−4.26	−4.32

Results are logarithmic to base 10.

of the following parameters:

- Time of escape T.
- The angle β of the escaping single body.
- The semi-major axis A of the final binary.
- The eccentricity E of the binary.
- The minimum perimeter P at the final triple approach.
- The moment of inertia I of the triple system at the same time.
- The minimum two-body distance R during the evolution (osculating value).
- The r.m.s. deviation ϵ between the initial coordinates of the bodies and the final coordinates after the reversed solution.
- The similar r.m.s. deviation ν for the velocities.

The average quantities and distributions for the different bodies m_1, m_2 and m_3 are in qualitative agreement with each other. On the basis of these results we make the following conclusions:

- The average escape time <T> for the predictable examples is equal to about 65 time units; its r.m.s. deviation is decreasing rapidly for the smallest deviation, $\delta_i = 5.0 \times 10^{-8}$. The upper boundary of T for these cases is equal to 120 time units. In comparison we note that the minimum value for the non-predictable cases is 60 time units; in the interval between these two values we find both predictable and non-predictable examples.
- The average perimeter <P> is about 1.5 distance units; therefore at the triple approach this value (sum of three distances) is about a factor 8 smaller than the initial perimeter.
- The average semi-major axis of the final binary is about 0.5, i.e. a factor 3 smaller than the perimeter of the average triple approach resulting in escape.
- The average eccentricities of the final binaries are rather large: $< E > = 0.85$, and about 60% of the binaries have eccentricities in the interval [0.9, 1.0]. The distribution of E is similar to the statistical distribution of Monaghan (1976) for the planar three-body problem and also agrees with results from other three-body and N-body simulations (see, e.g., Anosova and Orlov, 1985). The eccentricities tend to be smaller for variations of m_3; here the proportion of large eccentricities, $E > 0.9$, is about 50%. This is smaller than the fraction of such eccentricities for the variations of m_1 and m_2 (this value is about 70%).

TABLE VIII
Escaper statistics.

Escaper	N_{esc}	$< T_i >$	σ_T	$< P_i >$	σ_P	$< A_i >$	σ_A	$< E_i >$	σ_E
m_1	219	17.4	0.18	1.73	0.04	0.54	0.01	0.90	0.01
m_1	190	17.5	0.19	1.66	0.05	0.52	0.01	0.91	0.01
m_1	211	16.4	0.18	1.72	0.05	0.55	0.01	0.85	0.01
m_2	49	17.7	0.50	1.22	0.06	0.46	0.01	0.75	0.04
m_2	47	18.1	0.54	1.15	0.08	0.40	0.02	0.76	0.04
m_2	33	17.2	0.64	1.10	0.08	0.39	0.02	0.78	0.04
m_3	30	16.2	0.66	0.91	0.04	0.37	0.01	0.79	0.04
m_3	18	17.8	0.99	0.96	0.08	0.33	0.02	0.86	0.03
m_3	34	14.6	0.72	0.74	0.05	0.32	0.02	0.76	0.03

Escape times and dispersions are in units of the crossing time ($T_{cr} = 3.84$).

- We always observe some close two-body approaches of the bodies: $<R> = 1.0 \times 10^{-3.5}$ with an r.m.s. deviation of 0.5 for the exponent; almost in all cases two-body approaches with $R < 1.0 \times 10^{-3}$ are observed. In the case of close triple stars we may have a collision between two components (for a length unit of 2 AU), but in the case of wide triple systems (length unit of 2000 AU) we have only identified four cases of collisions (assuming solar radii). We can therefore distinguish two effects which may explain the absence of close non-hierarchical triple stars: fast escapes and/or fast merging of the components.
- The distributions of β were described above; here we note an increased tendency for $\beta = 70$ degrees when $\delta_i = 5.0 \times 10^{-8}$ (i.e. the asymptotic Pythagorean value). For larger δ_i the distribution of β is close to a uniform one; thus some effect of lost memory of initial conditions takes place.
- The uncertainties of the final results are rather smaller for variations of body m_3; in this case the escape times are also shorter. The uncertainties ϵ in the coordinates are about a factor of 5 greater than the velocity errors ν. This is due to the propagation of small velocity errors over many time units.

Table VIII shows some results for escape of bodies with different masses. The first three lines correspond to the escape of m_1 for initial deviations around each of the bodies and combined for all values of δ_i, followed by three lines each for m_2 and m_3, respectively. The total number of escapers, N_{esc}, is given in Column 2 for each set of experiments. Note that the average escape times and corresponding dispersions are now expressed in units of the crossing time ($T_{cr} = 3.84$). The percentage of escape for each component is 75, 15 and 10. These results are qualitatively in agreement with results for the general three-body problem; i.e. 80, 16 and 4 (see, e.g., Anosova and Orlov, 1985; Anosova, 1986). Averages and r.m.s.

deviations for the parameters T_i, P_i, A_i and E_i are also displayed in Table VIII. The parameters P_i and A_i depend slightly on which body has been varied and more strongly on the escaping single body. Thus the triple approach and final binary is closer when the intermediate and most massive body escape, which is consistent with theoretical expectations (i.e. energy requirements).

We have confirmed the result of Anosova and Zavalov (1981; 1989) and Anosova (1991) that escape generally occurs after a close triple approach of a definite type when the future escaper passes between the final binary components along an approximately straight-line orbit, nearly normal to the binary line of apsides. We have also confirmed a result of Agekian and Anosova (1991) that escape is more probable if the binary components are receding at the moment of the closest triple approach. More detailed analysis of the triple approaches will be presented elsewhere.

6. Conclusions

We have attempted a systematic investigation of initial conditions in the neighbourhood of the classical Pythagorean three-body problem. The solutions are characterized by regular regions surrounded by stochastic regions. In the latter cases it is not possible to obtain any reliable numerical solutions. Moreover, some regular solutions also exceed evolution times which can be time-reversed to satisfactory accuracy using standard precision (i.e. 16 significant figures). The latter problem is alleviated somewhat by using a criterion of conditional escape which shortens the time-scale for orbits that would otherwise be long-lived. However, there still remains a significant fraction of systems with satisfactory solutions, especially bearing in mind that the planar problem with zero initial velocities poses a particularly serious numerical challenge due to the tendency for close triple approaches. Since all general solutions appear to terminate by escape, the final outcome of the motion is regular in the asymptotic sense; therefore our dynamical system represents an example of 'order out of chaos' (Prigogine and Stengers, 1984).

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